

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 15 — Halogenoalkanes (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions & Detailed Solutions

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
COMPONENT	Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)
TOPIC COVERAGE	15.1 Nucleophilic Substitution: Mechanisms (SN1 vs SN2), Reagents & Reactivity Trends · 15.2 Elimination Reactions
TOTAL VOLUME	110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)

HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

SYLLABUS SUBTOPICS COVERED IN THIS PACK

- 15.1 Nucleophilic Substitution & Mechanisms** — Nucleophilic substitution by :OH⁻ (hydrolysis to alcohols), :CN⁻ (chain extension to nitriles), and :NH₃ (excess ethanol to form amines)
- 15.2 Elimination Reactions & Environmental Impact** — Elimination of hydrogen halides by heating with ethanolic KOH (base-promoted proton abstraction from beta-carbon)
- High-Frequency Core Repeats** — The 10 most frequently examined Cambridge Paper 1 questions on Halogenoalkanes (Q101 – Q110).

SUBTOPIC 15.1 — NUCLEOPHILIC SUBSTITUTION & MECHANISMS (Q1 – Q55)

1 [HARD] Relative Rates of Hydrolysis of Halogenoalkanes — 9701/11/M/J/23/Q20

[1]

Equal amounts of 1-chlorobutane, 1-bromobutane, and 1-iodobutane are added to separate test-tubes containing aqueous silver nitrate dissolved in ethanol at 50 °C.

Which statement correctly describes the order of precipitate formation and the underlying reason?

- [A] 1-chlorobutane reacts fastest because the C-Cl bond is the most polar and attracts nucleophiles most strongly.
- [B] 1-iodobutane reacts fastest because the C-I bond enthalpy is the lowest (240 kJ mol⁻¹), making the C-I bond easiest to break despite being the least polar.
- [C] 1-bromobutane reacts fastest because bromine is in the middle of Group 17.
- [D] All three react at the identical rate because silver ions precipitate all halides simultaneously.

2 [MEDIUM] SN1 vs SN2 Mechanism Classification — 9701/12/M/J/23/Q20

[1]

Which pair correctly identifies the predominant substitution mechanism for 1-bromobutane and 2-bromo-2-methylpropane respectively?

- [A] 1-bromobutane reacts via SN1; 2-bromo-2-methylpropane reacts via SN2.
- [B] Both compounds react exclusively via SN1.
- [C] Both compounds react exclusively via SN2.
- [D] 1-bromobutane reacts predominantly via SN2; 2-bromo-2-methylpropane reacts predominantly via SN1.

3 [HARD] Stereochemical Outcome of the SN2 Mechanism — 9701/13/M/J/23/Q20

[1]

When an optically active pure (S)-enantiomer of 2-bromobutane reacts with aqueous hydroxide ions via a pure SN2 mechanism, what is the stereochemical nature of the resulting butan-2-ol?

- [A] It undergoes complete inversion of configuration (Walden inversion) to yield pure (R)-butan-2-ol.
- [B] It forms an optically inactive 50:50 racemic mixture of (R) and (S) enantiomers.
- [C] It retains its exact original (S) configuration with no structural change.
- [D] It undergoes planar elimination to form achiral but-2-ene.

4 [EASY] Carbon Chain Extension using Potassium Cyanide — 9701/11/O/N/23/Q20

[1]

What reagent and conditions are used to convert bromoethane into propanenitrile, extending the carbon chain by one carbon atom?

- [A] Hydrogen cyanide gas (HCN) in aqueous acid at 0 °C.
- [B] Concentrated aqueous ammonia heated under high pressure.
- [C] Potassium cyanide (KCN) dissolved in ethanol, heated under reflux.
- [D] Dilute potassium hydroxide in ethanol at room temperature.

5 [EASY] Reaction of Haloalkanes with Ammonia — 9701/13/M/J/22/Q5

[1]

What is the product when 1-bromopropane is heated under pressure with an EXCESS of concentrated ethanolic ammonia?

- [A] Propan-1-amine, CH₃CH₂CH₂NH₂ (primary amine)
- [B] Dipropylamine (secondary amine)
- [C] Tripropylamine (tertiary amine)
- [D] Tetrapropylammonium bromide

6 [HARD] Formation of Quaternary Ammonium Salts — 9701/11/M/J/22/Q6

[1]

What product is obtained when bromoethane is heated with ammonia when bromoethane is present in EXCESS?

- [A] Pure ethylamine only.
- [B] Ethanenitrile and hydrogen bromide.
- [C] Ethane and nitrogen gas.
- [D] A mixture containing primary, secondary, and tertiary amines plus tetraethylammonium bromide.

7 [EASY] Hydrolysis of Haloalkanes using Aqueous Alkali — 9701/12/M/J/222/Q7

[1]

What are the reagents and conditions used to convert 1-chlorobutane into butan-1-ol?

- [A] Ethanolic potassium hydroxide, heated under reflux.
- [B] Aqueous sodium hydroxide (NaOH(aq)), heated under reflux.
- [C] Concentrated sulfuric acid at 170°C .
- [D] Sodium metal in dry ether.

8 [HARD] Role of Ethanol in Silver Nitrate Test — 9701/13/M/J/223/Q8

[1]

Why is ethanol added to the reaction mixture when testing the rate of hydrolysis of haloalkanes with aqueous silver nitrate?

- [A] Ethanol acts as the nucleophile that displaces the halogen.
- [B] Ethanol oxidizes the halide to halogen.
- [C] Ethanol acts as a mutual co-solvent to dissolve both the non-polar haloalkane and the aqueous silver nitrate into a single homogeneous solution.
- [D] Ethanol precipitates silver metal.

9 [EASY] Carbocation Intermediate in $\text{S}_{\text{N}}1$ — 9701/11/M/J/224/Q9

[1]

What is the geometry and hybridisation of the carbocation intermediate formed in the first step of an $\text{S}_{\text{N}}1$ reaction?

- [A] Tetrahedral with sp^3 hybridisation
- [B] Trigonal planar with sp^2 hybridisation (120° bond angles)
- [C] Linear with sp hybridisation
- [D] Trigonal pyramidal with sp^3 hybridisation

10 [HARD] Racemisation in $\text{S}_{\text{N}}1$ Mechanism — 9701/12/M/J/220/Q10

[1]

Why does the $\text{S}_{\text{N}}1$ hydrolysis of an optically active tertiary halogenoalkane produce an optically inactive racemic mixture?

- [A] The planar carbocation intermediate can be attacked by the nucleophile with equal 50% probability from either the top face or bottom face.
- [B] The reaction proceeds with inversion only.
- [C] The halogen atom cannot detach.
- [D] The product undergoes spontaneous ring closure.

11 [EASY] Bond Polarity vs Bond Enthalpy — 9701/13/M/J/221/Q11

[1]

The C-F bond is the most polar carbon-halogen bond (highest δ^+ on carbon). Why is fluoroethane practically inert to nucleophilic substitution at 100°C ?

- [A] Fluorine atoms repel nucleophiles by electrostatic charges.
- [B] Fluoroethane is a solid at 100°C .
- [C] Fluoride ions cannot leave under any conditions.
- [D] The C-F bond enthalpy is extraordinarily high (485 kJ mol^{-1}), presenting an insurmountable activation energy barrier.

12 [HARD] Order of $\text{S}_{\text{N}}1$ Reaction Kinetics — 9701/11/M/J/222/Q12

[1]

What is the kinetic rate equation for the alkaline hydrolysis of 2-bromo-2-methylpropane ($\text{S}_{\text{N}}1$)?

- [A] $\text{Rate} = k[(\text{CH}_3)_3\text{CBr}][\text{OH}^-]$
- [B] $\text{Rate} = k[\text{OH}^-]^2$
- [C] $\text{Rate} = k[(\text{CH}_3)_3\text{CBr}]$ (first-order kinetics, independent of $[\text{OH}^-]$)
- [D] $\text{Rate} = k[(\text{CH}_3)_3\text{CBr}]^2$

13 [EASY] Order of SN2 Reaction Kinetics — 9701/12/M/J/223/Q13

[1]

What is the kinetic rate equation for the hydrolysis of 1-bromobutane with aqueous NaOH (SN2)?

- [A] $\text{Rate} = k[\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}][\text{OH}^-]$ (second-order kinetics)
- [B] $\text{Rate} = k[\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}]$
- [C] $\text{Rate} = k[\text{OH}^-]$
- [D] $\text{Rate} = k[\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}]^2$

14 [HARD] Transition State of SN2 Mechanism — 9701/13/M/J/224/Q14

[1]

Which statement correctly describes the transition state of an SN2 nucleophilic substitution?

- [A] A stable tetrahedral intermediate with an octet of electrons.
- [B] A planar three-coordinate carbocation.
- [C] A five-coordinate species where the incoming nucleophile and the departing leaving group are partially bonded at 180° to the central carbon.
- [D] A free radical with an unpaired electron.

15 [EASY] Hydrolysis with Water Alone — 9701/11/M/J/220/Q15

[1]

Why does 2-chloro-2-methylpropane react with cold neutral water to form 2-methylpropan-2-ol, whereas 1-chlorobutane requires boiling with aqueous NaOH?

- [A] Water is a stronger nucleophile than hydroxide.
- [B] 2-chloro-2-methylpropane dissociates readily via an SN1 pathway to form a stable 3° carbocation that is rapidly attacked by neutral water molecules.
- [C] 1-chlorobutane is an acid.
- [D] Chlorine in 1-chlorobutane is locked by resonance.

16 [HARD] Cyanide Substitution Mechanism — 9701/12/M/J/221/Q16

[1]

In the reaction between 1-bromopropane and cyanide ions, which atom of the cyanide ion ($\text{:C}\equiv\text{N}^-$) attacks the δ^+ carbon atom?

- [A] The nitrogen atom, because it is more electronegative.
- [B] Both carbon and nitrogen attack simultaneously.
- [C] The triple bond attacks directly.
- [D] The carbon atom, because it bears the formal negative charge and non-bonding lone pair.

17 [EASY] Hydrolysis of Nitriles — 9701/13/M/J/222/Q17

[1]

What organic product is formed when propanenitrile is heated under reflux with dilute hydrochloric acid?

- [A] Propan-1-amine
- [B] Propanal
- [C] Propanoic acid ($\text{CH}_3\text{CH}_2\text{COOH}$) and ammonium chloride
- [D] Ethyl ethanoate

18 [HARD] Alkaline Hydrolysis of Nitriles — 9701/11/M/J/223/Q18

[1]

What product is formed when ethanenitrile is heated with aqueous sodium hydroxide?

- [A] Sodium ethanoate (CH_3COONa) and ammonia gas (NH_3)
- [B] Ethanoic acid and sodium nitrate
- [C] Ethylamine
- [D] Ethanol

19 [EASY] Reduction of Nitriles — 9701/12/M/J/224/Q19**[1]**

Which reagent reduces propanenitrile to propan-1-amine?

- [A] Sodium borohydride (NaBH_4) in water
- [B] Acidified potassium dichromate
- [C] Hot concentrated sulfuric acid
- [D] Lithium aluminium hydride (LiAlH_4) in dry ether (or H_2 with Ni catalyst)

20 [HARD] Silver Halide Precipitate Colors — 9701/13/M/J/220/Q20**[1]**

When aqueous silver nitrate is added to separate test tubes containing chloride, bromide, and iodide ions, what are the observed colors of the precipitates?

- [A] AgCl : yellow; AgBr : cream; AgI : white
- [B] AgCl : white; AgBr : cream; AgI : yellow
- [C] AgCl : white; AgBr : brown; AgI : black
- [D] AgCl : red; AgBr : white; AgI : yellow

21 [EASY] Solubility of Silver Halides in Ammonia — 9701/11/M/J/221/Q21**[1]**

How can precipitates of AgCl , AgBr , and AgI be distinguished using aqueous ammonia?

- [A] All three dissolve in dilute NH_3 .
- [B] AgCl dissolves in dilute NH_3 ; AgBr dissolves only in concentrated NH_3 ; AgI is insoluble in both dilute and concentrated NH_3 .
- [C] AgI dissolves in dilute NH_3 ; AgCl is insoluble.
- [D] None of the precipitates dissolve in ammonia.

22 [HARD] Steric Hindrance in $\text{S}_\text{N}2$ — 9701/12/M/J/222/Q22**[1]**

Why does neopentyl bromide, $(\text{CH}_3)_3\text{C}-\text{CH}_2\text{Br}$, react extremely slowly in $\text{S}_\text{N}2$ substitution despite being a primary halogenoalkane?

- [A] It forms an unstable carbocation.
- [B] The C-Br bond is exceptionally strong.
- [C] Neopentyl bromide is a gas at room temperature.
- [D] The three bulky methyl groups on the adjacent beta-carbon sterically block the path of the incoming nucleophile toward the alpha-carbon.

23 [EASY] Primary Haloalkanes and $\text{S}_\text{N}1$ — 9701/13/M/J/223/Q23**[1]**

Why do primary halogenoalkanes NOT react via the $\text{S}_\text{N}1$ pathway under ordinary conditions?

- [A] Primary carbocations ($\text{R}-\text{CH}_2^+$) are thermodynamically unstable due to lack of charge dispersal by inductive effects.
- [B] Primary halogenoalkanes have no leaving group.
- [C] Primary carbons are sp^2 hybridised.
- [D] Primary haloalkanes cannot dissolve in water.

24 [HARD] Secondary Haloalkanes Mechanism — 9701/11/M/J/224/Q24**[1]**

Which substitution mechanism is utilized by secondary halogenoalkanes, such as 2-bromopropane?

- [A] Exclusively $\text{S}_\text{N}1$ under all conditions.
- [B] Exclusively $\text{S}_\text{N}2$ under all conditions.
- [C] A mixture of both $\text{S}_\text{N}1$ and $\text{S}_\text{N}2$ mechanisms operating concurrently depending on solvent polarity and nucleophile strength.
- [D] Neither; secondary haloalkanes only undergo elimination.

25 [EASY] Solvent Effects on SN1 vs SN2 — 9701/12/M/J/220/Q25**[1]**

How does using a polar protic solvent (like water or ethanol) affect the rate of an SN1 reaction compared to a polar aprotic solvent?

- [A] Polar protic solvents accelerate SN1 by stabilizing the departing halide anion and carbocation intermediate through hydrogen bonding.
- [B] Polar protic solvents stop SN1 completely.
- [C] Polar protic solvents force SN2 mechanism.
- [D] Solvents have no effect on reaction rates.

26 [HARD] Leaving Group Ability Down Group 17 — 9701/13/M/J/221/Q26**[1]**

Why is iodide (I⁻) a better leaving group than chloride (Cl⁻)?

- [A] Iodide is more electronegative than chloride.
- [B] Iodide forms stronger bonds with carbon.
- [C] Iodide has fewer electrons than chloride.
- [D] Iodide is larger, holds its negative charge with lower charge density, and is a weaker base (conjugate base of strong acid HI).

27 [EASY] Effect of Nucleophile Concentration on SN1 — 9701/11/M/J/222/Q27**[1]**

If the concentration of sodium hydroxide is doubled in the hydrolysis of 2-bromo-2-methylpropane, what happens to the reaction rate?

- [A] The rate doubles.
- [B] The rate remains unchanged because the reaction is zero-order with respect to hydroxide (SN1).
- [C] The rate quadruples.
- [D] The rate is halved.

28 [HARD] Effect of Nucleophile Concentration on SN2 — 9701/12/M/J/223/Q28**[1]**

If the concentration of both 1-bromobutane and NaOH are doubled, what happens to the rate of the SN2 reaction?

- [A] The rate doubles.
- [B] The rate remains unchanged.
- [C] The reaction rate increases by a factor of 4.
- [D] The rate increases by a factor of 8.

29 [EASY] Preparation of Haloalkanes from Alcohols using PCl5 — 9701/13/M/J/224/Q29**[1]**

What is observed when solid phosphorus(V) chloride, PCl5, is added to an alcohol at room temperature?

- [A] A bright yellow precipitate is formed.
- [B] Steamy acidic fumes of hydrogen chloride gas (HCl) are evolved vigorously with a hissing sound.
- [C] The solution turns green.
- [D] Carbon dioxide gas is evolved.

30 [HARD] Preparation of Haloalkanes using SOCl2 — 9701/11/M/J/220/Q30**[1]**

Why is thionyl chloride, SOCl2, considered the ideal laboratory reagent for converting alcohols to chloroalkanes?

- [A] Both byproducts (SO2 and HCl) are gases that escape from the reaction mixture, leaving pure chloroalkane without complex purification.
- [B] It reacts only at -50 °C.
- [C] It does not require any glassware.
- [D] It produces solid gold byproducts.

31 [EASY] Preparation of Bromoalkanes using NaBr and H₂SO₄ — 9701/12/M/J/221/Q31

[1]

How is a bromoalkane prepared from an alcohol in the laboratory?

- [A] By bubbling bromine gas into the alcohol in sunlight.
- [B] By reacting with liquid bromine and iron filings.
- [C] By boiling with aqueous sodium bromide.
- [D] By warming the alcohol with solid sodium bromide and 50% concentrated sulfuric acid (which generates HBr in situ).

32 [HARD] Preparation of Iodoalkanes using Red Phosphorus and Iodine — 9701/13/M/J/222/Q32

[1]

How is an iodoalkane prepared from an alcohol?

- [A] By heating with concentrated sulfuric acid and potassium iodide.
- [B] By bubbling hydrogen iodide through water.
- [C] By heating the alcohol under reflux with a mixture of red phosphorus and iodine (which forms PI₃ in situ).
- [D] By adding aqueous silver iodide.

33 [EASY] Why H₂SO₄ Cannot Be Used with KI — 9701/11/M/J/223/Q33

[1]

Why can concentrated sulfuric acid NOT be used with potassium iodide to prepare iodoalkanes from alcohols?

- [A] Concentrated H₂SO₄ oxidizes the iodide ions to iodine (I₂) and toxic H₂S/SO₂ rather than producing HI.
- [B] KI decomposes violently on contact with water.
- [C] Iodoalkanes explode in sulfuric acid.
- [D] Potassium iodide is completely insoluble in all acids.

34 [HARD] Walden Inversion Umbrella Model — 9701/12/M/J/224/Q34

[1]

What physical analogy is universally used to describe the inversion of stereochemical configuration during an S_N2 reaction?

- [A] A spinning coin settling flat.
- [B] A lock and key mechanism.
- [C] An umbrella blowing inside out in a violent wind gust.
- [D] A pendulum swinging back and forth.

35 [EASY] Solvolysis Definition — 9701/13/M/J/220/Q35

[1]

What is solvolysis in organic chemistry?

- [A] The thermal decomposition of a solute.
- [B] A nucleophilic substitution reaction in which the solvent molecule itself acts as the attacking nucleophile.
- [C] The extraction of a solute using chromatography.
- [D] The freezing of a solvent.

36 [HARD] Reactivity: Chlorobenzene vs Chloroalkane — 9701/11/M/J/221/Q36

[1]

Why is chlorobenzene (C₆H₅Cl) extremely unreactive towards nucleophilic substitution compared to 1-chlorobutane?

- [A] The benzene ring repels all electrons.
- [B] Chlorobenzene contains no chlorine atoms.
- [C] Chlorobenzene has a lower boiling point.
- [D] A lone pair on the chlorine atom overlaps with the delocalised pi system of the benzene ring, giving the C-Cl bond partial double-bond character and strengthening it.

37 [EASY] Nucleophilicity of Halide Ions in Protic Solvents — 9701/12/M/J/222/Q37**[1]**

What is the order of nucleophilicity of halide ions in polar protic solvents (such as water)?

- [A] $F^- > Cl^- > Br^- > I^-$
- [B] $Cl^- > Br^- > I^- > F^-$
- [C] $I^- > Br^- > Cl^- > F^-$
- [D] All halide ions have identical nucleophilicity.

38 [HARD] Hydrolysis of Acyl Chlorides vs Chloroalkanes — 9701/13/M/J/223/Q38**[1]**

Why do acyl chlorides ($RCOCl$) hydrolyze violently with cold water while chloroalkanes require heating with $NaOH(aq)$?

- [A] The carbonyl carbon is attached to two highly electronegative atoms (O and Cl), making it intensely electron-deficient and reactive via addition-elimination.
- [B] Acyl chlorides are ionic salts.
- [C] Chloroalkanes are completely non-polar.
- [D] Acyl chlorides contain no covalent bonds.

39 [EASY] Primary vs Tertiary Haloalkane Hydrolysis Reaction — 9701/11/M/J/224/Q39**[1]**

When shaken with aqueous silver nitrate at room temperature, 2-chloro-2-methylpropane produces a white precipitate within 10 seconds, whereas 1-chlorobutane shows no precipitate after 10 minutes. What explains this difference?

- [A] 1-chlorobutane is insoluble in ethanol.
- [B] Tertiary haloalkanes are strong Bronsted acids.
- [C] Chloride ions are absent in 1-chlorobutane.
- [D] 2-chloro-2-methylpropane ionizes rapidly via an S_N1 pathway to yield a stable tertiary carbocation, whereas 1-chlorobutane requires high activation energy for S_N2 attack by water.

40 [HARD] Haloalkane Boiling Point Trend — 9701/12/M/J/220/Q40**[1]**

Why does 1-iodobutane have a higher boiling point ($130\text{ }^\circ\text{C}$) than 1-chlorobutane ($78\text{ }^\circ\text{C}$)?

- [A] The C-I bond is much more polar.
- [B] Iodine has significantly more electrons, resulting in a larger and more polarizable electron cloud that generates stronger London dispersion forces.
- [C] 1-iodobutane forms hydrogen bonds.
- [D] 1-chlorobutane is more volatile due to ring strain.

41 [EASY] Haloalkane Density Trend — 9701/13/M/J/221/Q41**[1]**

Why are liquid bromoalkanes and iodoalkanes denser than water, whereas liquid alkanes and monochloroalkanes are less dense than water?

- [A] Haloalkanes contain heavy ionic crystals.
- [B] The high atomic mass of bromine (79.9) and iodine (126.9) adds substantial mass per unit molecular volume.
- [C] Water pushes haloalkanes down by magnetic repulsion.
- [D] Haloalkanes react with water to form heavy salts.

42 [HARD] Identification of Halogen from Mass Spec Isotope Peaks — 9701/11/M/J/222/Q42**[1]**

A halogenoalkane shows two molecular ion peaks of virtually equal intensity at $m/z = 108$ and $m/z = 110$ ($M : M+2 = 1 : 1$). Which halogen atom is present?

- [A] One chlorine atom
- [B] Two chlorine atoms
- [C] One iodine atom
- [D] One bromine atom (due to ^{79}Br and ^{81}Br in a 50.7 : 49.3 natural abundance ratio)

43 [EASY] Chlorine Isotope Ratio in Mass Spec — 9701/12/M/J/223/Q43

[1]

A chloroalkane exhibits an $[M]^+$ peak at $m/z = 78$ and an $[M+2]^+$ peak at $m/z = 80$ in a 3:1 height ratio. What is the reason for this ratio?

- [A] Chlorine exists naturally as 75% ^{35}Cl and 25% ^{37}Cl isotopes.
- [B] Carbon-13 is present in 33% abundance.
- [C] The molecule fragments 75% of the time.
- [D] Deuterium causes the $M+2$ peak.

44 [HARD] Refluxing Haloalkane with Ethanolic KCN Mechanism — 9701/13/M/J/224/Q44

[1]

In the reaction $\text{CH}_3\text{CH}_2\text{Br} + \text{KCN} \rightarrow \text{CH}_3\text{CH}_2\text{CN} + \text{KBr}$, what type of bond fission and mechanism take place?

- [A] Homolytic fission via free radical substitution
- [B] Electrophilic addition across the cyano group
- [C] Heterolytic fission of the C-Br bond via nucleophilic substitution
- [D] Elimination of HBr followed by addition

45 [EASY] Di-haloalkane Hydrolysis — 9701/11/M/J/220/Q45

[1]

What is the product when 1,2-dibromoethane is heated under reflux with excess aqueous sodium hydroxide?

- [A] Ethane-1,2-diol (ethylene glycol), $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$
- [B] Ethanal
- [C] Ethanoic acid
- [D] Ethene

46 [HARD] Gem-dihaloalkane Hydrolysis — 9701/12/M/J/221/Q46

[1]

What organic product is formed when 1,1-dichloroethane is heated with aqueous NaOH?

- [A] Ethanol
- [B] Ethanoic acid
- [C] Ethane-1,2-diol
- [D] Ethanal, CH_3CHO

47 [EASY] Tri-haloalkane Hydrolysis — 9701/13/M/J/222/Q47

[1]

What organic compound is formed upon complete alkaline hydrolysis of 1,1,1-trichloroethane followed by acidification?

- [A] Ethanal
- [B] Ethanoic acid, CH_3COOH
- [C] Ethanol
- [D] Trichloromethane

48 [HARD] Reactivity of Fluoroalkanes in Industry — 9701/11/M/J/223/Q48

[1]

Why are fluoroalkanes chosen for artificial joints and non-stick coatings?

- [A] Fluoroalkanes dissolve easily in body fluids.
- [B] Fluoroalkanes act as biological catalysts.
- [C] The extreme strength of the C-F bond makes them completely biologically inert, heat-resistant, and non-toxic.
- [D] Fluoroalkanes conduct electricity efficiently.

49 [EASY] KCN Reaction Safety Precautions — 9701/12/M/J/224/Q49**[1]**

Why must the reaction of haloalkanes with potassium cyanide be performed in a dedicated fume cupboard?

- [A] Cyanide gas burns with an invisible flame.
- [B] Any accidental acidification of cyanide salts releases deadly, volatile hydrogen cyanide gas (HCN), which inhibits cellular respiration.
- [C] KCN is an explosive solid.
- [D] Ethanol reacts with KCN to produce mustard gas.

50 [HARD] Synthesis of Amines via Nitrile Reduction vs Ammonia Substitution — 9701/13/M/J/220/Q50**[1]**

Why is the reduction of nitriles preferred over the reaction of haloalkanes with ammonia for preparing pure primary amines?

- [A] Nitrile reduction yields pure primary amine with zero contamination by secondary or tertiary amines.
- [B] Nitrile reduction does not require any heat.
- [C] Haloalkane reaction with ammonia is endothermic.
- [D] Nitriles are completely non-toxic.

51 [EASY] Distinguishing 1-bromobutane and 2-bromo-2-methylpropane — 9701/11/M/J/221/Q51**[1]**

Which test reagent can immediately differentiate 1-bromobutane from 2-bromo-2-methylpropane at room temperature within 30 seconds?

- [A] Aqueous sodium carbonate.
- [B] Bromine water.
- [C] Acidified potassium manganate(VII).
- [D] Aqueous silver nitrate in ethanol (2-bromo-2-methylpropane forms a cream precipitate rapidly via S_N1 ; 1-bromobutane shows no immediate precipitate).

52 [HARD] S_N2 Transition State Inversion Proof — 9701/12/M/J/222/Q52**[1]**

How was the Walden inversion mechanism historically proven by chemists?

- [A] By observing the molecule under an optical microscope.
- [B] By measuring the freezing point of the mixture.
- [C] By reacting an optically active haloalkane with radioactive labeled halide ($*X^-$) and demonstrating that the rate of loss of optical activity was exactly twice the rate of radioactive exchange.
- [D] By combusting the product in pure oxygen.

53 [EASY] Carbocation Rearrangement in Haloalkane Hydrolysis — 9701/13/M/J/223/Q53**[1]**

When 1-bromo-2,2-dimethylpropane is hydrolyzed under conditions that promote carbocation formation, why is 2-methylbutan-2-ol the primary product?

- [A] The initial 1° carbocation undergoes a 1,2-methyl shift to generate a stable 3° carbocation before reacting with water.
- [B] Water attacks the methyl groups directly.
- [C] The molecule undergoes ring expansion.
- [D] Bromine abstracts a proton from the solvent.

54 [HARD] Reaction Coordinate Diagram of S_N1 vs S_N2 — 9701/11/M/J/224/Q54**[1]**

How do the reaction coordinate energy profiles of S_N1 and S_N2 reactions differ fundamentally?

- [A] S_N1 is a flat line.
- [B] S_N2 has three intermediates.
- [C] S_N1 has two energy peaks (two transition states) separated by a central carbocation energy well (intermediate); S_N2 has a single energy peak (one transition state) with no intermediate.
- [D] S_N1 has only one transition state.

55 [EASY] Solvent Dielectric Constant Effect — 9701/12/M/J/220/Q55**[1]**

How does increasing the solvent dielectric constant (polarity) influence the rate of S_N1 substitution reactions?

- [A] It decreases the rate because solvent molecules collide with reactants.
- [B] It increases the reaction rate substantially by lowering the Gibbs free energy of activation for heterolytic bond dissociation.
- [C] It converts all S_N1 reactions into free radical substitutions.
- [D] It has zero effect on the rate constant.

SUBTOPIC 15.2 — ELIMINATION REACTIONS & ENVIRONMENTAL IMPACT (Q56 – Q100)**56** [EASY] Elimination Reaction of Halogenoalkanes — 9701/11/M/J/23/Q21**[1]**

What reagents and conditions are required to convert 2-bromopropane into propene, and what is the role of the hydroxide ion?

- [A] Aqueous sodium hydroxide at room temperature; hydroxide acts as a nucleophile.
- [B] Concentrated sulfuric acid at 170 °C; sulfuric acid acts as an oxidizing agent.
- [C] Potassium cyanide in ethanol; cyanide acts as a catalyst.
- [D] Ethanolic potassium hydroxide (KOH in ethanol), heated under reflux; hydroxide acts as a Bronsted-Lowry base.

57 [HARD] Regiochemistry in Elimination (Zaitsev's Rule) — 9701/12/M/J/23/Q21**[1]**

When 2-bromobutane is heated with ethanolic potassium hydroxide, a mixture of three isomeric alkenes is formed.

Which alkene is the major product and why?

- [A] But-1-ene; primary hydrogens are less sterically hindered and abstract faster.
- [B] Cis-but-2-ene; cis isomers are always thermodynamically more stable.
- [C] Trans-but-2-ene; it is the most highly substituted and sterically stable alkene formed via elimination of hydrogen from C3.
- [D] 2-methylpropene; the carbon skeleton undergoes rearrangement.

58 [HARD] CFCs and Catalytic Ozone Depletion in the Stratosphere — 9701/13/M/J/23/Q21**[1]**

In the upper stratosphere, chlorofluorocarbons such as CCl₃F (CFC-11) break down under solar UV radiation to catalyze ozone depletion.

Which reaction represents the initiation step, and why is a chlorine radical formed rather than a fluorine radical?

- [A] CCl₃F → ·CCl₂F + Cl·; the C-Cl bond enthalpy (340 kJ mol⁻¹) is lower than the C-F bond enthalpy (485 kJ mol⁻¹), so UV photons selectively cleave the C-Cl bond.
- [B] CCl₃F → ·CCl₃ + F·; fluorine is more electronegative and leaves first.
- [C] CCl₃F + O₃ → ClO· + CO₂ + F₂
- [D] CCl₃F → CCl₂ + ClF (molecular elimination)

59 [HARD] Catalytic Ozone Depletion Cycle Equations — 9701/11/O/N/23/Q21**[1]**

Which pair of equations correctly represents the homogeneous catalytic cycle by which chlorine free radicals destroy ozone in the stratosphere?

- [A] Step 1: Cl· + O₂ → ClO₂·; Step 2: ClO₂· + O₃ → Cl· + 2O₂
- [B] Step 1: 2Cl· + O₃ → Cl₂O + O₂; Step 2: Cl₂O → 2Cl· + O
- [C] Step 1: Cl· + O₃ → ClO₃·; Step 2: ClO₃· → Cl· + 1.5 O₂
- [D] Step 1: Cl· + O₃ → ClO· + O₂; Step 2: ClO· + O → Cl· + O₂ (Net: O₃ + O → 2O₂)

60 [HARD] Hydrofluoroalkanes as CFC Replacements — 9701/11/M/J/220/Q60**[1]**

Why are hydrofluoroalkanes (HFAs, e.g. CH₂FCF₃) used as environmentally safe replacements for CFCs in aerosol propellants and air conditioners?

- [A] They have zero global warming potential.
- [B] They contain no C-Cl bonds, so they cannot produce ozone-destroying chlorine radicals in the stratosphere.
- [C] They are solid polymers that cannot evaporate.
- [D] They react with ozone to form beneficial oxygen.

61 [EASY] Elimination vs Substitution Solvent Control — 9701/12/M/J/221/Q61**[1]**

How can the reaction between 2-bromobutane and potassium hydroxide be directed to yield butan-2-ol rather than butene isomers?

- [A] By using ethanolic potassium hydroxide heated under reflux.
- [B] By using aqueous potassium hydroxide at moderate temperature.
- [C] By adding concentrated sulfuric acid.
- [D] By carrying out the reaction in pure liquid ammonia.

62 [HARD] Greenhouse Effect of CFCs and HFAs — 9701/13/M/J/222/Q62**[1]**

Although hydrofluoroalkanes (HFAs) do not deplete the ozone layer, what other major environmental concern is associated with their release?

- [A] They dissolve in rain to form concentrated hydrofluoric acid.
- [B] They cause toxic smog in the mesosphere.
- [C] They deplete oxygen from the oceans.
- [D] They are potent greenhouse gases with high global warming potentials (GWPs) that absorb infrared radiation strongly.

63 [EASY] Elimination of 1-bromobutane — 9701/11/M/J/223/Q63**[1]**

What is the only alkene formed when 1-bromobutane is heated with ethanolic potassium hydroxide?

- [A] But-1-ene, $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$
- [B] But-2-ene
- [C] 2-methylpropene
- [D] Cyclobutane

64 [HARD] Elimination of 2-bromo-2-methylbutane — 9701/12/M/J/224/Q64**[1]**

When 2-bromo-2-methylbutane is heated with ethanolic KOH, what are the two possible alkene products?

- [A] Pent-1-ene and pent-2-ene
- [B] 3-methylbut-1-ene and 2-methylbut-2-ene
- [C] 2-methylbut-2-ene (major) and 2-methylbut-1-ene (minor)
- [D] 2,3-dimethylbut-2-ene only

65 [EASY] Inertness of CFCs in Troposphere — 9701/13/M/J/220/Q65**[1]**

Why do chlorofluorocarbons (CFCs) survive unchanged in the lower atmosphere (troposphere) for over 50 to 100 years?

- [A] They are non-flammable, insoluble in water, and resistant to oxidation by hydroxyl radicals ($\cdot\text{OH}$) due to high bond enthalpies.
- [B] They are absorbed by soil bacteria.
- [C] They react with nitrogen gas to form stable complexes.
- [D] They freeze into ice crystals in clouds.

66 [HARD] Free Radical Homolysis of Halogenoalkanes — 9701/11/M/J/221/Q66**[1]**

What wavelength of ultraviolet light is required to cleave the C-Cl bond in CFC molecules in the stratosphere?

- [A] Visible light (400–700 nm)
- [B] Infrared radiation (1000 nm)
- [C] Radio waves
- [D] Short-wavelength UV-C radiation (wavelength < 240 nm)

67 [EASY] Role of Ozone in the Stratosphere — 9701/12/M/J/222/Q67**[1]**

Why is the stratospheric ozone layer essential for protecting terrestrial life on Earth?

- [A] It prevents heat from escaping the Earth's core.
- [B] It absorbs harmful high-energy UV-B and UV-C radiation that causes skin cancer, cataracts, and DNA mutations.
- [C] It maintains atmospheric oxygen at 21%.
- [D] It neutralizes acid rain before it falls.

68 [HARD] Elimination with Bulky Bases — 9701/13/M/J/223/Q68**[1]**

How does using a bulky, sterically hindered base (like potassium tert-butoxide) alter the regiochemical outcome of elimination in 2-bromobutane?

- [A] It produces 100% trans-but-2-ene.
- [B] It stops elimination completely and causes substitution.
- [C] It favors the less substituted alkene, but-1-ene (Hofmann product), because the bulky base cannot easily access the more crowded secondary hydrogens.
- [D] It isomerises but-2-ene into cyclobutane.

69 [EASY] Mechanism of E2 Elimination — 9701/11/M/J/224/Q69**[1]**

What is the molecularity and timing of bond changes in an E2 elimination mechanism?

- [A] Unimolecular: carbocation forms first followed by slow proton loss.
- [B] Bimolecular and concerted: base removes the proton while the C=C double bond forms and the leaving group departs simultaneously in one step.
- [C] Bimolecular: leaving group leaves, then base attacks 10 minutes later.
- [D] Free radical addition.

70 [HARD] Mechanism of E1 Elimination — 9701/12/M/J/220/Q70**[1]**

Under what conditions does an E1 elimination occur, and what intermediate is formed?

- [A] Tertiary haloalkanes in polar protic solvents; a carbocation intermediate is formed in the slow step, followed by proton loss to the solvent.
- [B] Primary haloalkanes with concentrated base.
- [C] Gas-phase reactions under UV light.
- [D] Reactions involving free radicals only.

71 [EASY] Anti-periplanar Geometry in E2 — 9701/13/M/J/221/Q71**[1]**

Why must the beta-hydrogen and the leaving group halogen atom adopt an anti-periplanar (180°) geometry in an E2 transition state?

- [A] It minimizes molecular weight.
- [B] It prevents collisions with the solvent.
- [C] It maximizes dipole attraction between H and X.
- [D] This orientation allows optimal parallel alignment of the breaking C-H and C-X sigma orbitals to overlap directly into the new pi bond.

72 [HARD] Montreal Protocol Significance — 9701/11/M/J/222/Q72**[1]**

What was the primary objective and global outcome of the 1987 Montreal Protocol international environmental treaty?

- [A] To ban the use of all plastics globally.
- [B] To tax carbon dioxide emissions from power stations.
- [C] To phase out the global production and consumption of ozone-depleting substances, successfully halting the expansion of the Antarctic ozone hole.
- [D] To eliminate the use of internal combustion engines.

73 [EASY] Nitrogen Oxides in Ozone Depletion — 9701/12/M/J/223/Q73**[1]**

Besides chlorine radicals from CFCs, which other free radical catalyst contributes significantly to stratospheric ozone depletion?

- [A] Nitrogen monoxide radicals (NO) emitted by high-altitude supersonic aircraft engines.
- [B] Methane molecules.
- [C] Carbon dioxide molecules.
- [D] Argon atoms.

74 [HARD] Natural Formation of Ozone (Chapman Cycle) — 9701/13/M/J/224/Q74**[1]**

How is ozone naturally formed in the stratosphere from oxygen gas?

- [A] Oxygen burns in volcanic gases.
- [B] Lightning in the mesosphere fuses nitrogen and oxygen.
- [C] UV-C photolysis splits O_2 into oxygen atoms ($\text{O}_2 \rightarrow 2\text{O}$), which combine with undecomposed O_2 molecules: $\text{O} + \text{O}_2 \rightarrow \text{O}_3$.
- [D] Plants release ozone directly during photosynthesis.

75 [EASY] Chlorine Radical Lifetime in Stratosphere — 9701/11/M/J/220/Q75**[1]**

Why can a single chlorine atom destroy up to 100,000 ozone molecules before being deactivated?

- [A] Chlorine atoms never react with anything except ozone.
- [B] The chlorine radical is regenerated in each catalytic cycle and is only removed when it eventually reacts with methane or NO_2 to form stable reservoir molecules.
- [C] Chlorine atoms replicate inside ozone clouds.
- [D] Chlorine atoms are radioactive.

76 [HARD] Reservoir Molecules for Stratospheric Chlorine — 9701/12/M/J/221/Q76**[1]**

What are the two major inactive 'reservoir' molecules that temporarily store chlorine in the stratosphere?

- [A] Dichlorine monoxide (Cl_2O) and chlorine gas (Cl_2)
- [B] Sodium chloride and potassium chlorate
- [C] Chloromethane and chloroform
- [D] Hydrogen chloride (HCl) and chlorine nitrate (ClONO_2)

77 [EASY] Polar Stratospheric Clouds (PSCs) — 9701/13/M/J/222/Q77**[1]**

What role do Polar Stratospheric Clouds (PSCs) play in the annual springtime formation of the Antarctic ozone hole?

- [A] PSCs freeze ozone into liquid drops that fall to Earth.
- [B] PSCs produce massive amounts of sulfur dioxide.
- [C] Heterogeneous reactions on ice crystals in PSCs convert inactive reservoir species (HCl and ClONO_2) into active molecular chlorine (Cl_2), which is photolyzed to $\cdot\text{Cl}$ by spring sunlight.
- [D] PSCs absorb all UV radiation.

78 [HARD] HCFCs as Transitional Substances — 9701/11/M/J/223/Q78**[1]**

Why were hydrochlorofluorocarbons (HCFCs, e.g. CHClF_2) used only as temporary transitional substitutes rather than permanent replacements for CFCs?

- [A] They still contain a C-Cl bond and have a non-zero (though lower) ozone depletion potential.
- [B] They are too expensive to synthesize.
- [C] They are highly explosive at room temperature.
- [D] They turn into toxic sulfuric acid.

79 [EASY] Deducing Alkene Products from Di-haloalkane Elimination — 9701/12/M/J/224/Q79

[1]

What product is formed when 1,2-dibromopropane is heated with an excess of ethanolic potassium hydroxide?

- [A] Propene
- [B] Cyclopropane
- [C] Propane-1,2-diol
- [D] Propyne, $\text{CH}\equiv\text{C}-\text{CH}_3$ (alkyne)

80 [HARD] Elimination in Cyclic Halides — 9701/13/M/J/220/Q80

[1]

What is the product when bromocyclohexane is heated under reflux with ethanolic KOH?

- [A] Cyclohexanol
- [B] Cyclohexene
- [C] Benzene
- [D] Hex-1-ene

81 [EASY] Reactivity of Halogenoalkanes with Magnesium — 9701/11/M/J/221/Q81

[1]

What useful organometallic synthetic reagent is formed when 1-bromobutane reacts with magnesium metal in dry diethyl ether?

- [A] Butane and magnesium bromide
- [B] Butylmagnesium bromide, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{MgBr}$ (a Grignard reagent)
- [C] Octane and magnesium hydride
- [D] But-1-ene and magnesium

82 [HARD] Wurtz Coupling Reaction — 9701/12/M/J/222/Q82

[1]

What hydrocarbon is produced when 1-bromoethane is heated with sodium metal in dry ether?

- [A] Ethane
- [B] Ethene
- [C] Hexane
- [D] Butane, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$

83 [EASY] Dehydrohalogenation Temperature Comparison — 9701/13/M/J/223/Q83

[1]

Why does the dehydrohalogenation (elimination) of 2-bromobutane require higher temperatures than the nucleophilic substitution of the same compound?

- [A] Elimination has a higher activation energy because it involves breaking both a C-H bond and a C-Br bond simultaneously.
- [B] Elimination is an endothermic reaction.
- [C] Substitution requires breaking four bonds.
- [D] Base molecules move slower than nucleophiles.

84 [HARD] Distinguishing Elimination from Substitution Experimentally — 9701/11/M/J/224/Q84

[1]

How can an experimenter verify that elimination rather than substitution took place when 2-chlorobutane was treated with KOH?

- [A] Test the liquid product with acidified silver nitrate.
- [B] Measure the pH with universal indicator.
- [C] Test the gaseous product with bromine water; immediate decolorization confirms the formation of an unsaturated alkene.
- [D] Test for carbon dioxide with limewater.

85 [EASY] Halon Fire Extinguishers — 9701/12/M/J/220/Q85**[1]**

What are Halons (e.g. CBrClF₂), and why was their use in building fire suppression systems banned under environmental regulations?

- [A] Brominated fluorocarbons used as fire extinguishants; bromine radicals released in the stratosphere are 40–100 times more destructive to ozone than chlorine radicals.
- [B] They explode upon contact with oxygen.
- [C] They form poisonous cyanide gas in fires.
- [D] They react with nitrogen to produce nitric acid.

86 [HARD] Toxicity of Chlorinated Solvents — 9701/13/M/J/221/Q86**[1]**

Why has the industrial use of trichloroethene and 1,1,1-trichloroethane as degreasing solvents been strictly curtailed?

- [A] They are insoluble in all organic liquids.
- [B] They decompose into pure hydrochloric acid at room temperature.
- [C] They are powerful oxidizers that burn in air.
- [D] They are volatile organic pollutants that cause liver toxicity, central nervous system depression, and carcinogenicity.

87 [EASY] Elimination of 3-bromopentane — 9701/11/M/J/222/Q87**[1]**

When 3-bromopentane is heated with ethanolic potassium hydroxide, what alkene is formed?

- [A] Pent-1-ene
- [B] Pent-2-ene, CH₃-CH=CH-CH₂-CH₃
- [C] 2-methylbut-2-ene
- [D] Cyclopentane

88 [HARD] Effect of Alkyl Branching on Elimination Rate — 9701/12/M/J/223/Q88**[1]**

How does the rate of elimination vary among primary, secondary, and tertiary halogenoalkanes under identical basic conditions?

- [A] Primary > Secondary > Tertiary
- [B] All three undergo elimination at identical rates.
- [C] Tertiary > Secondary > Primary
- [D] Secondary > Primary > Tertiary

89 [EASY] Elimination with Alcoholic Base vs Aqueous Base — 9701/13/M/J/224/Q89**[1]**

Why does an alcoholic solution of KOH contain a higher concentration of the stronger ethoxide base (CH₃CH₂O⁻) than an aqueous solution?

- [A] Ethanol oxidizes hydroxide to ethoxide.
- [B] Ethanol is a weaker acid than water, so equilibrium establishes the strongly basic ethoxide ion: OH⁻ + CH₃CH₂OH ⇌ H₂O + CH₃CH₂O⁻.
- [C] Water destroys all basic species.
- [D] Potassium reacts with ethanol to evolve hydrogen.

90 [HARD] Structure of Ozone Molecule — 9701/11/M/J/220/Q90**[1]**

What is the shape and bonding in an ozone (O₃) molecule?

- [A] Bent (angular) with an O-O-O bond angle of ~117° and delocalised pi bonding across the three oxygen atoms.
- [B] Linear with 180° bond angle.
- [C] Equilateral triangular ring.
- [D] Trigonal planar with 120° bond angle.

91 [EASY] Role of Sunlight in Ozone Hole Recovery — 9701/12/M/J/221/Q91**[1]**

Why does the Antarctic ozone hole appear specifically in the austral spring (September–October) rather than in mid–winter?

- [A] Antarctica is warmer in winter than in spring.
- [B] Ozone only freezes in summer.
- [C] Wind patterns push ozone away during spring.
- [D] Solar UV radiation is required to photolyse accumulated Cl_2 and HOCl into reactive chlorine radicals ($\cdot\text{Cl}$) to trigger the catalytic chain.

92 [HARD] Comparison of C–X Bond Enthalpies — 9701/13/M/J/222/Q92**[1]**

Which sequence ranks carbon–halogen bond dissociation enthalpies in order of DECREASING bond strength?

- [A] $\text{C–I} > \text{C–Br} > \text{C–Cl} > \text{C–F}$
- [B] $\text{C–Cl} > \text{C–F} > \text{C–Br} > \text{C–I}$
- [C] $\text{C–F} (485 \text{ kJ mol}^{-1}) > \text{C–Cl} (340 \text{ kJ mol}^{-1}) > \text{C–Br} (280 \text{ kJ mol}^{-1}) > \text{C–I} (240 \text{ kJ mol}^{-1})$
- [D] $\text{C–Br} > \text{C–Cl} > \text{C–I} > \text{C–F}$

93 [EASY] Elimination Mechanism Transition State Orbital Overlap — 9701/11/M/J/223/Q93**[1]**

In the E_2 transition state, what type of orbital is formed as the C–H and C–X sigma bonds break?

- [A] A pi (π) bond formed by the lateral overlap of developing parallel 2p orbitals on the two adjacent carbons.
- [B] A new sigma bond formed along the internuclear axis.
- [C] An sp^3 hybrid orbital.
- [D] A coordinate dative bond to the base.

94 [HARD] Elimination Products of 1-bromo-2-methylpropane — 9701/12/M/J/224/Q94**[1]**

What is the sole alkene produced when 1-bromo-2-methylpropane is heated with ethanolic KOH?

- [A] But-1-ene
- [B] But-2-ene
- [C] 2-methylpropene, $(\text{CH}_3)_2\text{C}=\text{CH}_2$
- [D] Methylcyclopropane

95 [EASY] Atmospheric Residence Time of CFC-12 — 9701/13/M/J/220/Q95**[1]**

Why does dichlorodifluoromethane (CFC-12) have an atmospheric lifetime exceeding 100 years?

- [A] It reacts with nitrogen to form permanent rocks.
- [B] It does not undergo photolysis in the troposphere, is totally insoluble in rainwater, and diffuses very slowly into the high stratosphere.
- [C] It is heavier than all air molecules and sinks into deep caves.
- [D] It is magnetic and bound to the poles.

96 [HARD] Kinetics of Elimination vs Substitution — 9701/11/M/J/221/Q96**[1]**

How does increasing the concentration of base affect the competition between $\text{S}_\text{N}2$ substitution and E_2 elimination?

- [A] It causes $\text{S}_\text{N}1$ to dominate.
- [B] It stops both reactions.
- [C] It selectively accelerates only substitution.
- [D] Both rates increase proportionally, but higher base concentration and higher temperature shift the product ratio toward elimination.

97 [EASY] Elimination of 2-chloropentane — 9701/12/M/J/222/Q97**[1]**

What are the three alkene stereoisomers produced when 2-chloropentane is heated with ethanolic KOH?

- [A] Pent-1-ene and pent-2-ene only
- [B] 2-methylbut-1-ene, 2-methylbut-2-ene, and pent-1-ene
- [C] Pent-1-ene, cis-pent-2-ene, and trans-pent-2-ene
- [D] Cyclopentane, pent-1-ene, and pent-2-ene

98 [HARD] Global Warming Potential of CFCs — 9701/13/M/J/223/Q98**[1]**

Why is the Global Warming Potential (GWP) of CFC-11 approximately 5,000 times greater than that of CO₂ over a 100-year timescale?

- [A]** CFC-11 absorbs thermal infrared radiation strongly in the atmospheric window region (8–14 micrometers) where CO₂ and H₂O do not absorb.
- [B]** CFC-11 produces 5,000 molecules of CO₂ when it breaks down.
- [C]** CFC-11 emits heat through nuclear decay.
- [D]** CFC-11 reflects sunlight back to space.

99 [EASY] Environmental Safer Alternative - HFOs — 9701/11/M/J/224/Q99**[1]**

What are Hydrofluoroolefins (HFOs, such as CF₃-CH=CH₂) and why are they considered superior fourth-generation refrigerants?

- [A]** They are non-flammable noble gases.
- [B]** They are permanently frozen at room temperature.
- [C]** They are made directly from carbon dioxide.
- [D]** They contain a reactive C=C double bond that allows them to be destroyed rapidly in the troposphere by .OH radicals, resulting in zero ODP and near-zero GWP.

100 [HARD] Chemical Test for Distinguishing Haloalkane from Alkane — 9701/12/M/J/220/Q100**[1]**

Which sequence of tests confirms that a colorless liquid is a chloroalkane rather than an alkane?

- [A]** Add bromine water in the dark.
- [B]** Reflux with aqueous NaOH, acidify with dilute HNO₃, then add aqueous AgNO₃; a white precipitate confirms the presence of chloride.
- [C]** Shake with cold acidified potassium manganate(VII).
- [D]** Test with damp red litmus paper.

FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)**101 [HARD] [HIGH-FREQUENCY EXAM REPEAT]** HF 1: Deducing the Number of Alkene Isomers from Dehydrohalogenation — 9701/11/M/J/22/Q20**[1]**

When 2-bromobutane is heated under reflux with ethanolic potassium hydroxide, a mixture of isomeric alkenes is produced.

Including both structural isomers and stereoisomers (cis-trans), how many distinct alkene products are formed?

- [A]** 2 alkenes
- [B]** 4 alkenes
- [C]** 3 alkenes (but-1-ene, cis-but-2-ene, and trans-but-2-ene)
- [D]** 1 alkene

102 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 2: Relative Rate of Precipitate Formation with Ethanolic Silver Nitrate — 9701/12/M/J/21/Q20**[1]**

Four separate test tubes contain 1-chlorobutane, 1-bromobutane, 1-iodobutane, and 2-chloro-2-methylpropane. Equal volumes of ethanolic silver nitrate are added to each tube at 50 °C.

Which compound produces a precipitate most rapidly, and which produces a precipitate most slowly?

- [A]** Fastest: 2-chloro-2-methylpropane (due to rapid S_N1 ionization); Slowest: 1-chlorobutane (due to strong C-Cl bond and slow S_N2).
- [B]** Fastest: 1-chlorobutane; Slowest: 1-iodobutane.
- [C]** Fastest: 1-iodobutane; Slowest: 2-chloro-2-methylpropane.
- [D]** Fastest: 1-bromobutane; Slowest: 1-chlorobutane.

103 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 3: Multi-Step Synthesis Involving Chain Extension via Nitrile — 9701/13/O/N/22/Q20**[1]**

Consider the following two-step organic synthetic conversion:
 $\text{CH}_3\text{CH}_2\text{Br} \xrightarrow{\text{Step 1}} \text{X} \xrightarrow{\text{Step 2}} \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$

What are the reagents and conditions for Step 1 and Step 2?

- [A] Step 1: NH_3 in ethanol, heated under pressure; Step 2: H_2 with Ni catalyst.
- [B] Step 1: NaOH(aq) , heated; Step 2: NH_3 under pressure.
- [C] Step 1: KCN in dilute acid; Step 2: NaBH_4 in water.
- [D] Step 1: KCN in ethanol, heated under reflux; Step 2: LiAlH_4 in dry ether (or H_2 with Ni catalyst).

104 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 4: Identifying $\text{S}_\text{N}1$ vs $\text{S}_\text{N}2$ by Kinetic Rate Laws — 9701/11/F/M/22/Q20**[1]**

The rate of hydrolysis of two halogenoalkanes, P and Q, was investigated:

- For compound P, doubling the concentration of sodium hydroxide doubled the rate of reaction.
- For compound Q, doubling the concentration of sodium hydroxide had no effect on the rate of reaction.

What structural classes do P and Q belong to?

- [A] P is a tertiary halogenoalkane; Q is a primary halogenoalkane.
- [B] P is a primary halogenoalkane (reacting via $\text{S}_\text{N}2$); Q is a tertiary halogenoalkane (reacting via $\text{S}_\text{N}1$).
- [C] Both P and Q are secondary halogenoalkanes.
- [D] P is an alkene; Q is an alkane.

105 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 5: Catalytic Chain Mechanism of Ozone Depletion by Chlorine Radicals — 9701/12/O/N/21/Q21**[1]**

A single chlorine free radical released in the stratosphere can destroy tens of thousands of ozone molecules before being deactivated.

Why does the chlorine radical have such an immense catalytic turnover number?

- [A] It is consumed in the first propagation step ($\text{Cl} \cdot + \text{O}_3 \rightarrow \text{ClO} \cdot + \text{O}_2$) but regenerated in the second propagation step ($\text{ClO} \cdot + \text{O} \rightarrow \text{Cl} \cdot + \text{O}_2$), remaining free to attack further ozone molecules.
- [B] It multiplies by splitting into two radicals in each step.
- [C] It is completely unreactive towards all other stratospheric gases.
- [D] It acts as a solid heterogeneous catalyst surface.

106 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 6: Stereochemical Inversion in the $\text{S}_\text{N}2$ Hydrolysis of Chiral Halogenoalkanes — 9701/13/M/J/22/Q20**[1]**

A sample of pure (R)-2-chloropentane is hydrolyzed by heating with concentrated aqueous sodium hydroxide under conditions where the reaction proceeds exclusively via an $\text{S}_\text{N}2$ mechanism.

What is the optical nature of the resulting pentan-2-ol product?

- [A] A racemic mixture of (R)- and (S)-pentan-2-ol, which is optically inactive.
- [B] Pure (R)-pentan-2-ol, showing retention of configuration.
- [C] Pure (S)-pentan-2-ol, showing optical activity due to complete inversion of configuration (Walden inversion).
- [D] An optically inactive meso compound.

107 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 7: Distinguishing Halogenoalkanes by Reaction with Ethanolic Ammonia — 9701/11/O/N/21/Q20**[1]**

When 1-bromobutane is heated with concentrated ethanolic ammonia in a sealed tube, which set of organic species is present in the final equilibrium mixture?

- [A] Butylamine only
- [B] Butylamine, dibutylamine, tributylamine, and tetrabutylammonium bromide
- [C] Butanenitrile and ammonium bromide
- [D] But-1-ene and ammonium bromide

108 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 8: Environmental Chemistry — Ozone Depletion Potential Comparison — 9701/12/F/M/23/Q21 [1]

Which compound has an Ozone Depletion Potential (ODP) of ZERO?

- [A] Dichlorodifluoromethane, CCl_2F_2 (CFC-12)
- [B] Trichlorofluoromethane, CCl_3F (CFC-11)
- [C] Bromotrifluoromethane, CBrF_3 (Halon-1301)
- [D] 1,1,1,2-tetrafluoroethane, $\text{CF}_3\text{-CH}_2\text{F}$ (HFC-134a)

109 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 9: Predicting Elimination Products of Symmetrical Di-haloalkanes — 9701/13/M/J/23/Q21 [1]

2,3-dibromobutane is treated with excess ethanolic potassium hydroxide under reflux.

What is the final stable organic hydrocarbon product formed after complete dehydrohalogenation?

- [A] But-2-yne, $\text{CH}_3\text{-C}\equiv\text{C-CH}_3$
- [B] Buta-1,3-diene
- [C] Cyclobutene
- [D] Butane

110 [HARD] [HIGH-FREQUENCY EXAM REPEAT] HF 10: Quantitative Titration Analysis of Halide Hydrolysis — 9701/11/M/J/23/Q22 [1]

0.010 mol of an unknown chloroalkane is completely hydrolyzed by boiling with aqueous sodium hydroxide. The mixture is acidified with dilute nitric acid, and excess aqueous silver nitrate is added.

The resulting dry white precipitate has a mass of 2.87 g.

How many chlorine atoms are present in one molecule of the chloroalkane? (Ar: Ag = 107.9, Cl = 35.5)

- [A] 1 chlorine atom
- [B] 3 chlorine atoms
- [C] 2 chlorine atoms (it is a dichloroalkane)
- [D] 4 chlorine atoms

QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

Q1 – Q20		Q21 – Q40		Q41 – Q60		Q61 – Q80		Q81 – Q100	
Q01	B	Q21	B	Q41	B	Q61	B	Q81	B
Q02	D	Q22	D	Q42	D	Q62	D	Q82	D
Q03	A	Q23	A	Q43	A	Q63	A	Q83	A
Q04	C	Q24	C	Q44	C	Q64	C	Q84	C
Q05	A	Q25	A	Q45	A	Q65	A	Q85	A
Q06	D	Q26	D	Q46	D	Q66	D	Q86	D
Q07	B	Q27	B	Q47	B	Q67	B	Q87	B
Q08	C	Q28	C	Q48	C	Q68	C	Q88	C
Q09	B	Q29	B	Q49	B	Q69	B	Q89	B
Q10	A	Q30	A	Q50	A	Q70	A	Q90	A
Q11	D	Q31	D	Q51	D	Q71	D	Q91	D
Q12	C	Q32	C	Q52	C	Q72	C	Q92	C
Q13	A	Q33	A	Q53	A	Q73	A	Q93	A
Q14	C	Q34	C	Q54	C	Q74	C	Q94	C
Q15	B	Q35	B	Q55	B	Q75	B	Q95	B
Q16	D	Q36	D	Q56	D	Q76	D	Q96	D
Q17	C	Q37	C	Q57	C	Q77	C	Q97	C
Q18	A	Q38	A	Q58	A	Q78	A	Q98	A
Q19	D	Q39	D	Q59	D	Q79	D	Q99	D
Q20	B	Q40	B	Q60	B	Q80	B	Q100	B

FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

Q101 C	Q102 A	Q103 D	Q104 B	Q105 A	Q106 C	Q107 B	Q108 D	Q109 A	Q110 C
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COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
1	B	Option B is correct. The rate-determining factor in the hydrolysis of halogenoalkanes is the carbon-halogen bond dissociation enthalpy: C-Cl (340 kJ/mol) > C-Br (280 kJ/mol) > C-I (240 kJ/mol). Even though C-Cl has the highest dipole moment, bond strength dominates. The weaker C-I bond breaks most readily, liberating iodide ions that instantly form a yellow AgI precipitate first.	[1]
2	D	Option D is correct. Primary halogenoalkanes (like 1-bromobutane) have minimal steric hindrance around the alpha-carbon, allowing backside attack by the nucleophile in a concerted one-step SN2 mechanism. Tertiary halogenoalkanes (like 2-bromo-2-methylpropane) suffer severe steric crowding preventing backside attack, but can form a stable tertiary carbocation stabilized by three alkyl groups (+I effect), reacting via the two-step SN1 pathway.	[1]
3	A	Option A is correct. In an SN2 mechanism, the incoming nucleophile (:OH-) must attack the carbon atom from the face exactly opposite to the departing bromide leaving group (180° backside attack). As the new C-O bond forms and C-Br breaks, the three other substituent groups invert like an umbrella blowing inside out in a gale (Walden inversion), completely inverting the optical configuration.	[1]
4	C	Option C is correct. Heating a halogenoalkane with ethanolic KCN (or NaCN) under reflux causes nucleophilic substitution: CH3CH2Br + :CN- → CH3CH2C#N + Br-. The cyanide ion donates a lone pair to form a new C-C bond, successfully lengthening the carbon backbone by one carbon atom.	[1]
5	A	Option A is correct. Using a large excess of ethanolic ammonia ensures that an unreacted NH3 molecule is far more likely to attack the haloalkane than the newly formed amine, suppressing further alkylation and maximizing the yield of the primary amine propan-1-amine.	[1]
6	D	Option D is correct. With excess halogenoalkane, the primary amine formed retains a lone pair and acts as a nucleophile, reacting successively to form diethylamine, triethylamine, and finally the quaternary ammonium salt (C2H5)4N+ Br-.	[1]
7	B	Option B is correct. Nucleophilic substitution to form an alcohol requires aqueous alkali (NaOH(aq) or KOH(aq)) heated under reflux. Water provides a polar ionizing medium that favors substitution over elimination.	[1]
8	C	Option C is correct. Haloalkanes are insoluble in water and would form a separate immiscible layer. Ethanol is miscible with both water and organic halogenoalkanes, enabling effective molecular collisions between reactants in a homogeneous single phase.	[1]
9	B	Option B is correct. In the slow step of SN1, the leaving group departs with its electron pair, leaving a trivalent carbon with six valence electrons in three sp2 hybrid orbitals, forming a planar intermediate with 120° angles and an empty unhybridised p orbital.	[1]
10	A	Option A is correct. Because the carbocation intermediate is completely planar, the incoming nucleophile has an identical 50% chance of attacking from above or below the plane, producing an equimolar mixture of both enantiomers (racemate), resulting in zero net optical rotation.	[1]
11	D	Option D is correct. Although carbon has a high delta+ charge, substitution requires breaking the C-X bond. The C-F bond is extremely short and strong (485 kJ/mol), making the activation energy for cleavage prohibitively high under ordinary chemical conditions.	[1]
12	C	Option C is correct. In the SN1 mechanism, the rate-determining step is the unimolecular heterolytic dissociation of the C-Br bond. The nucleophile (:OH-) participates only in the fast second step, so the rate law is first order overall: Rate = k[substrate].	[1]
13	A	Option A is correct. The SN2 mechanism involves a single concerted bimolecular step in which both the haloalkane and the hydroxide ion participate in the transition state. Thus, the rate is proportional to both concentrations: Rate = k[substrate][OH-].	[1]
14	C	Option C is correct. The SN2 transition state is a transient, unstable five-coordinate arrangement where the C-Nu bond is partially formed and the C-X bond is partially broken, with both groups aligned at 180° along the reaction axis carrying partial negative charges.	[1]
15	B	Option B is correct. The stability of the tertiary carbocation allows 2-chloro-2-methylpropane to ionize spontaneously in polar water. Water acts as a weak nucleophile (:OH2) to capture the carbocation. Primary haloalkanes cannot ionize and need the stronger nucleophile :OH- with heat.	[1]
16	D	Option D is correct. Cyanide (:C#N-) has a lone pair on both carbon and nitrogen, but the carbon atom is less electronegative and more polarizable, carrying the negative charge. Nucleophilic attack occurs via the carbon atom, forming a stable carbon-carbon bond (nitrile).	[1]
17	C	Option C is correct. Acid hydrolysis of nitriles converts the -C#N group into a carboxylic acid: CH3CH2CN + HCl + 2H2O → CH3CH2COOH + NH4Cl.	[1]
18	A	Option A is correct. Refluxing a nitrile with aqueous alkali produces a carboxylate salt and evolves ammonia gas: CH3CN + NaOH + H2O → CH3COONa + NH3.	[1]
19	D	Option D is correct. Nitriles are reduced to primary amines using the powerful reducing agent LiAlH4: CH3CH2CN + 4[H] → CH3CH2CH2NH2 (or catalytic hydrogenation with H2/Ni).	[1]
20	B	Option B is correct. Precipitation of silver halides gives distinctive colors: AgCl is a white precipitate; AgBr is a cream (pale yellow) precipitate; AgI is a bright yellow precipitate.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
21	B	Option B is correct. The stability of the diamminesilver(I) complex $[\text{Ag}(\text{NH}_3)_2]^+$ allows AgCl to dissolve in dilute NH_3 . The lower solubility product (K_{sp}) of AgBr requires concentrated NH_3 to dissolve, while AgI has such a tiny K_{sp} that it remains insoluble even in concentrated NH_3 .	[1]
22	D	Option D is correct. Although the alpha-carbon has two hydrogens (1°), the adjacent quaternary carbon bears three bulky methyl groups that physically obstruct the backside approach of nucleophiles, rendering $\text{S}_{\text{N}}2$ attack extremely slow.	[1]
23	A	Option A is correct. A primary carbocation has only one alkyl group donating electron density (+I effect). Its energy of formation is too high, making the activation energy barrier for unimolecular ionization prohibitively large.	[1]
24	C	Option C is correct. Secondary haloalkanes have intermediate steric hindrance and intermediate carbocation stability, allowing both $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ pathways to operate simultaneously; strong nucleophiles and non-polar solvents favor $\text{S}_{\text{N}}2$, while weak nucleophiles and polar protic solvents favor $\text{S}_{\text{N}}1$.	[1]
25	A	Option A is correct. In $\text{S}_{\text{N}}1$, the rate-determining step forms separated positive (R^+) and negative (X^-) ions. Polar protic solvents solvate both ions via dipole interactions and hydrogen bonds, drastically lowering the activation energy barrier.	[1]
26	D	Option D is correct. Good leaving groups are weak bases that stably accommodate electron density. Because HI is a stronger acid than HCl , I^- is a weaker base with a larger ionic radius (diffuse charge), making it leave much more readily than Cl^- .	[1]
27	B	Option B is correct. The rate law for $\text{S}_{\text{N}}1$ is $\text{Rate} = k[\text{substrate}]$. Hydroxide ion does not participate in the rate-determining step, so changing $[\text{OH}^-]$ has zero effect on the overall rate.	[1]
28	C	Option C is correct. The rate law for $\text{S}_{\text{N}}2$ is $\text{Rate} = k[\text{RBr}][\text{OH}^-]$. Doubling both concentrations multiplies the rate by $2 \times 2 = 4$.	[1]
29	B	Option B is correct. PCl_5 reacts vigorously with alcohols at room temperature: $\text{ROH} + \text{PCl}_5 \rightarrow \text{RCl} + \text{POCl}_3 + \text{HCl}(\text{g})$. Steamy acidic fumes of HCl gas are evolved, turning damp blue litmus paper red and forming dense white fumes with ammonia.	[1]
30	A	Option A is correct. Reaction: $\text{ROH} + \text{SOCl}_2 \rightarrow \text{RCl} + \text{SO}_2(\text{g}) + \text{HCl}(\text{g})$. Because both sulfur dioxide and hydrogen chloride are gases that bubble out of solution, the chloroalkane is obtained directly in high purity.	[1]
31	D	Option D is correct. HBr is generated in situ by reacting NaBr with 50% H_2SO_4 : $\text{NaBr} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HBr}$. The HBr then protonates the alcohol, followed by nucleophilic substitution by Br^- to yield the bromoalkane.	[1]
32	C	Option C is correct. PI_3 is unstable, so it is generated in situ by refluxing red phosphorus and iodine: $2\text{P} + 3\text{I}_2 \rightarrow 2\text{PI}_3$. PI_3 reacts with the alcohol: $3\text{ROH} + \text{PI}_3 \rightarrow 3\text{RI} + \text{H}_3\text{PO}_3$.	[1]
33	A	Option A is correct. Iodide is a powerful reducing agent. Concentrated H_2SO_4 oxidizes HI to elemental iodine (I_2) while sulfur is reduced to SO_2 , S , and H_2S . Instead, non-oxidizing phosphoric(V) acid (H_3PO_4) must be used.	[1]
34	C	Option C is correct. As the nucleophile attacks from the back and the leaving group departs from the front, the three non-reacting groups are forced through a planar transition state, inverting their spatial geometry like an umbrella turning inside out in the wind.	[1]
35	B	Option B is correct. Solvolysis refers to a substitution reaction where the solvent (e.g. H_2O in hydrolysis, or ethanol in ethanolysis) serves directly as the nucleophile.	[1]
36	D	Option D is correct. The p orbital lone pair of chlorine delocalises into the aromatic pi system, creating partial double bond character in the $\text{C}-\text{Cl}$ bond. This makes the bond shorter, much stronger (400 kJ/mol), and resistant to nucleophilic cleavage.	[1]
37	C	Option C is correct. In protic solvents, small ions with high charge density (F^-) are heavily solvated by strong hydrogen bonds, which must be broken before attack. The larger, more polarizable I^- ion is weakly solvated and donates its electrons readily, making it the most nucleophilic halide.	[1]
38	A	Option A is correct. In an acyl chloride ($\text{R}-\text{C}(=\text{O})\text{Cl}$), both the carbonyl oxygen and chlorine pull electron density away from the carbonyl carbon. This massive δ^+ charge attracts weak nucleophiles like water vigorously at room temperature via an addition-elimination mechanism.	[1]
39	D	Option D is correct. The tertiary haloalkane rapidly undergoes unimolecular heterolysis to form a stable 3° carbocation, immediately releasing free Cl^- ions that precipitate AgCl . The primary haloalkane cannot ionize spontaneously and water is too weak a nucleophile to displace chloride via $\text{S}_{\text{N}}2$ quickly at room temperature.	[1]
40	B	Option B is correct. Iodine has atomic number 53 (53 e^-) compared to chlorine (17 e^-). The much larger electron cloud is easily distorted, creating stronger temporary dipoles and London dispersion forces that require more thermal energy to separate molecules.	[1]
41	B	Option B is correct. Bromine and iodine atoms have very high atomic masses packed into a single substituent, significantly raising the mass-to-volume ratio (density $> 1.2 - 1.9 \text{ g/cm}^3$) well above water (1.0 g/cm^3).	[1]
42	D	Option D is correct. Bromine exists naturally as two isotopes, ^{79}Br (50.7%) and ^{81}Br (49.3%), with an approximate 1:1 abundance. A compound with one Br atom displays twin molecular ion peaks $[\text{M}]^+$ and $[\text{M}+2]^+$ of equal height separated by 2 mass units.	[1]
43	A	Option A is correct. Naturally occurring chlorine consists of ~75% ^{35}Cl and ~25% ^{37}Cl (a 3:1 ratio). Thus, any mono-chloro organic compound displays an $[\text{M}]^+$ and $[\text{M}+2]^+$ peak pattern with a characteristic 3:1 intensity ratio.	[1]
44	C	Option C is correct. The cyanide nucleophile attacks the δ^+ carbon, causing heterolytic fission of the polar $\text{C}-\text{Br}$ bond where both bonding electrons leave with the bromide ion: a nucleophilic substitution.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
45	A	Option A is correct. Both bromine atoms are primary and undergo nucleophilic substitution by :OH ⁻ to yield the vicinal diol ethane-1,2-diol (used extensively as engine antifreeze).	[1]
46	D	Option D is correct. Hydrolysis of 1,1-dichloroethane initially yields an unstable gem-diol [CH ₃ CH(OH) ₂], which spontaneously eliminates a molecule of water to produce the carbonyl compound ethanal (CH ₃ CHO).	[1]
47	B	Option B is correct. Substitution of all three chlorine atoms by -OH produces an ortho-acid intermediate [CH ₃ C(OH) ₃], which loses water to form the carboxylic acid ethanoic acid (CH ₃ COOH).	[1]
48	C	Option C is correct. The C-F bond is among the strongest known in chemistry (485 kJ/mol). Fluoroalkanes resist degradation by body enzymes, acids, bases, and oxygen, providing ideal inert materials for implants and prosthetics.	[1]
49	B	Option B is correct. Cyanide ions (:CN ⁻) are extremely toxic. If exposed to acidic conditions or breathed as vapor, lethal HCN gas is generated, which rapidly binds to cytochrome c oxidase in mitochondria, halting ATP production.	[1]
50	A	Option A is correct. Reaction of haloalkanes with ammonia produces a complex mixture of 1°, 2°, 3° amines and quaternary salts that are difficult to separate. In contrast, reduction of a pure nitrile (R-CN + 4[H] → R-CH ₂ NH ₂) gives exclusively the primary amine in high yield.	[1]
51	D	Option D is correct. At room temperature, the tertiary haloalkane rapidly ionizes (S _N 1) to liberate bromide ions, yielding a visible cream precipitate of AgBr almost immediately. The primary haloalkane reacts via S _N 2 far too slowly at room temperature to form a precipitate.	[1]
52	C	Option C is correct. Each S _N 2 substitution with radioactive *X ⁻ inverts one molecule from (+) to (-). Converting one (+) molecule to (-) removes two units of net optical rotation. Thus, the rate of loss of optical rotation was precisely twice the rate of chemical substitution, proving every single collision causes inversion.	[1]
53	A	Option A is correct. Ionization gives the neopentyl carbocation (CH ₃) ₃ C-CH ₂ ⁺ (1°). A methyl group with its bonding pair immediately migrates (1,2-methyl shift) to form the tertiary carbocation (CH ₃) ₂ C ⁺ -CH ₂ CH ₃ , which captures water to give the tertiary alcohol 2-methylbutan-2-ol.	[1]
54	C	Option C is correct. S _N 1 is a two-step mechanism: peak 1 (C-X bond breaking), intermediate well (carbocation), peak 2 (nucleophile attack). S _N 2 is a single concerted step with one transition state peak and no intermediate.	[1]
55	B	Option B is correct. Polar solvents with high dielectric constants stabilize the developing charges in the transition state and the ionic intermediate, significantly lowering the activation energy barrier for the rate-determining ionization step.	[1]
56	D	Option D is correct. When heated under reflux with ethanolic KOH (where ethanol is the solvent rather than water), the hydroxide ion acts as a base, abstracting a proton (H ⁺) from a beta-carbon atom. Concurrently, the bromide leaving group departs, forming a C=C double bond (propene + KBr + H ₂ O): an elimination reaction.	[1]
57	C	Option C is correct. Proton abstraction can occur from C1 (forming but-1-ene) or from C3 (forming but-2-ene). By Zaitsev's rule, elimination preferentially yields the more highly substituted alkene (but-2-ene), which is thermodynamically more stable. Furthermore, trans-but-2-ene minimizes steric repulsion between the two methyl groups compared to cis-but-2-ene, making trans-but-2-ene the predominant product.	[1]
58	A	Option A is correct. Solar UV radiation in the stratosphere (wavelength ~175-220 nm) possesses sufficient quantum energy to homolytically cleave the weaker C-Cl bond (340 kJ/mol) but cannot break the much stronger C-F bond (485 kJ/mol). This selectively generates chlorine free radicals (.Cl), which catalyze ozone destruction.	[1]
59	D	Option D is correct. The chlorine radical acts as a homogeneous catalyst: (1) Cl. abstracts an oxygen atom from ozone, forming a chlorine monoxide radical and O ₂ (Cl. + O ₃ → ClO. + O ₂); (2) ClO. reacts with a free oxygen atom (formed by UV photolysis of O ₂) to regenerate the chlorine radical and produce O ₂ (ClO. + O → Cl. + O ₂). The chlorine radical emerges unchanged to repeat the destruction cycle up to 100,000 times.	[1]
60	B	Option B is correct. HFAs contain only carbon, hydrogen, and fluorine. Because they lack chlorine atoms, photolysis in the stratosphere cannot generate chlorine radicals (.Cl), resulting in an Ozone Depletion Potential (ODP) of zero.	[1]
61	B	Option B is correct. Aqueous solvent solvates hydroxide ions with a hydration shell, reducing base strength and favoring nucleophilic substitution (forming butan-2-ol). Ethanolic solvent provides an ethoxide/hydroxide mixture that acts as a strong base, favoring elimination.	[1]
62	D	Option D is correct. Both CFCs and replacement HFAs possess strong C-F bonds that absorb infrared radiation intensely in the atmospheric window, making them greenhouse gases thousands of times more potent per molecule than CO ₂ .	[1]
63	A	Option A is correct. In 1-bromobutane (CH ₃ CH ₂ CH ₂ CH ₂ Br), beta-hydrogens are present only on Carbon-2. Abstraction of a proton from C2 with loss of Br ⁻ from C1 can only yield but-1-ene.	[1]
64	C	Option C is correct. Elimination can abstract a proton from the adjacent -CH ₃ groups (yielding 2-methylbut-1-ene) or from the -CH ₂ - group (yielding 2-methylbut-2-ene). By Zaitsev's rule, the more substituted alkene 2-methylbut-2-ene is the major product.	[1]
65	A	Option A is correct. CFCs possess only strong C-F and C-Cl bonds and lack reactive C-H or C=C bonds. They do not react with tropospheric oxidants like .OH radicals, nor do they dissolve in rain, allowing them to diffuse slowly into the stratosphere.	[1]
66	D	Option D is correct. The C-Cl bond enthalpy (340 kJ/mol) requires high-energy ultraviolet photons in the UV-C band (< 240 nm), which only penetrate into the upper stratosphere and are filtered out before reaching the Earth's surface.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
67	B	Option B is correct. Ozone (O ₃) absorbs solar ultraviolet radiation between 200 and 315 nm through photolysis (O ₃ + UV → O ₂ + O), shielding living organisms from destructive ionizing UV rays.	[1]
68	C	Option C is correct. Bulky bases like (CH ₃) ₃ CO ⁻ K ⁺ cannot easily reach the sterically crowded internal beta-protons at C3. Instead, they abstract the readily accessible, less hindered primary protons at C1, favoring but-1-ene.	[1]
69	B	Option B is correct. In the E2 (Elimination Bimolecular) mechanism, base attack on the beta-proton, C=C pi bond formation, and C-X cleavage occur concurrently in a single transition state with anti-periplanar geometry.	[1]
70	A	Option A is correct. E1 (Elimination Unimolecular) occurs in two steps: (1) rate-determining loss of leaving group to form a carbocation intermediate; (2) rapid loss of a beta-proton to a weak base/solvent to form the alkene. Favored by 3° substrates.	[1]
71	D	Option D is correct. In the anti-periplanar arrangement, the bonding electrons from the breaking C-H bond can seamlessly flow into the developing C=C pi orbital as the C-X antibonding orbital is pushed away, providing lowest activation energy.	[1]
72	C	Option C is correct. The Montreal Protocol was a landmark international treaty that mandated the total phase-out of CFCs, halons, and carbon tetrachloride, allowing the stratospheric ozone layer to begin measurable recovery.	[1]
73	A	Option A is correct. Nitrogen monoxide (.NO) from high-altitude aircraft exhausts catalyzes ozone destruction: NO. + O ₃ → NO ₂ . + O ₂ ; NO ₂ . + O → NO. + O ₂ ; net: O ₃ + O → 2O ₂ .	[1]
74	C	Option C is correct. The Chapman mechanism: intense solar UV (< 240 nm) dissociates molecular oxygen into reactive oxygen atoms (O ₂ → 2O). An oxygen atom then collides with an O ₂ molecule in the presence of a third body (M) to form ozone: O + O ₂ + M → O ₃ + M.	[1]
75	B	Option B is correct. Chlorine radicals act as true catalysts, emerging undamaged from each destruction cycle. They only terminate when they encounter trace species like methane (.Cl + CH ₄ → HCl + .CH ₃) or NO ₂ (.ClO + NO ₂ → ClONO ₂), forming temporary inert reservoir species.	[1]
76	D	Option D is correct. Chlorine radicals are temporarily deactivated when converted into stable reservoir molecules: hydrogen chloride (HCl, formed via .Cl + CH ₄) and chlorine nitrate (ClONO ₂ , formed via ClO. + NO ₂).	[1]
77	C	Option C is correct. During the dark Antarctic winter, PSCs of nitric acid trihydrate and water ice catalyze surface reactions: HCl + ClONO ₂ → Cl ₂ + HNO ₃ . When spring sunlight returns, Cl ₂ is instantly cleaved by sunlight into massive bursts of .Cl radicals, triggering runaway ozone loss.	[1]
78	A	Option A is correct. HCFCs contain C-H bonds that allow most molecules to be destroyed by .OH in the troposphere. However, a small fraction still reaches the stratosphere, where their C-Cl bonds release chlorine radicals, giving them a non-zero ODP.	[1]
79	D	Option D is correct. 1,2-dibromopropane has two bromine atoms on adjacent carbons. Heating with excess ethanolic KOH causes two successive elimination reactions (loss of two HBr molecules), introducing a triple bond to form the alkyne propyne.	[1]
80	B	Option B is correct. Elimination of HBr from bromocyclohexane removes the bromine atom and a neighboring ring proton, creating a carbon-carbon double bond to form cyclohexene.	[1]
81	B	Option B is correct. Refluxing a haloalkane with magnesium turnings in anhydrous ether inserts the Mg atom into the C-Br bond, forming an alkylmagnesium halide (Grignard reagent, R-Mg-X), where carbon carries a nucleophilic carbanion character (C ^{delta-}).	[1]
82	D	Option D is correct. The Wurtz reaction couples two alkyl radicals via sodium metal: 2CH ₃ CH ₂ Br + 2Na → CH ₃ CH ₂ CH ₂ CH ₃ + 2NaBr, producing butane.	[1]
83	A	Option A is correct. The simultaneous or sequential cleavage of a strong C-H bond (413 kJ/mol) alongside the C-Br bond in elimination demands a higher kinetic energy barrier, making high temperatures favorable for elimination over substitution.	[1]
84	C	Option C is correct. Substitution yields butan-2-ol (saturated alcohol, no reaction with bromine water); elimination yields butene isomers (gaseous alkenes), which bubble out and immediately decolorize bromine water from orange-brown to colorless.	[1]
85	A	Option A is correct. Halons contain bromine atoms (C-Br bonds). When photolyzed in the stratosphere, bromine radicals (.Br) destroy ozone with catalytic efficiencies 40 to 100 times higher than chlorine, resulting in massive ozone depletion potentials (ODPs up to 10 ⁻¹⁶).	[1]
86	D	Option D is correct. Chlorinated hydrocarbons are liver and kidney toxins, penetrate groundwater supplies due to high stability, and contribute to tropospheric VOC emissions and ozone layer depletion.	[1]
87	B	Option B is correct. In 3-bromopentane (CH ₃ CH ₂ CH(Br)CH ₂ CH ₃), both adjacent positions (C2 and C4) are identical -CH ₂ -methylene groups. Elimination from either side yields pent-2-ene (existing as cis and trans stereoisomers).	[1]
88	C	Option C is correct. Tertiary halogenoalkanes have more beta-hydrogens available for abstraction, form more stable, more highly substituted alkenes, and can proceed via lower-energy E1 or E2 pathways, reacting fastest in elimination.	[1]
89	B	Option B is correct. Because ethanol is less acidic than water, dissolving KOH in ethanol establishes an equilibrium containing ethoxide ions (C ₂ H ₅ O ⁻), which are less solvated and significantly stronger Bronsted bases than aqueous hydroxide, powerfully promoting proton abstraction (elimination).	[1]
90	A	Option A is correct. Ozone has 18 valence electrons. The central oxygen has one lone pair, one single coordinate bond, and one double bond, exhibiting resonance with a bent shape (~117° angle) and partial double-bond character in both O-O bonds.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
91	D	Option D is correct. During the pitch-black polar winter, cold temperatures form PSCs and build up Cl ₂ , but no sunlight is present. As the sun rises in the Antarctic spring, UV photons cleave Cl ₂ molecules into floods of reactive .Cl radicals, causing rapid, catastrophic ozone depletion.	[1]
92	C	Option C is correct. Descending Group 17, halogen atomic radius increases and the bonding orbital overlap with carbon becomes longer and more diffuse, resulting in a steady decrease in bond enthalpy: C-F (485) > C-Cl (340) > C-Br (280) > C-I (240 kJ/mol).	[1]
93	A	Option A is correct. As the base abstracts the beta-proton and the halogen leaves, the sp ³ orbitals on both carbons rehybridise toward sp ² , and their unhybridised 2p lobes overlap sideways to form the new C=C pi bond.	[1]
94	C	Option C is correct. In 1-bromo-2-methylpropane ((CH ₃) ₂ CH-CH ₂ Br), there is only one beta-hydrogen (on C2). Elimination of H from C2 and Br from C1 forms 2-methylpropene as the only possible alkene.	[1]
95	B	Option B is correct. CFC-12 contains no C-H bonds for .OH abstraction and no C=C bonds for addition. It cannot be washed out by precipitation and only breaks down when it reaches the high stratosphere above the ozone layer where intense UV-C exists, giving a residence time of ~100 years.	[1]
96	D	Option D is correct. Both S _N 2 and E2 are bimolecular (second order) processes depending on [base]. However, because elimination has a higher activation energy and entropy of activation, increasing temperature and using high base concentrations in non-aqueous solvents strongly favors E2 elimination over S _N 2 substitution.	[1]
97	C	Option C is correct. Abstraction from C1 gives pent-1-ene. Abstraction from C3 gives pent-2-ene, which exists as a pair of geometric stereoisomers: cis-pent-2-ene and trans-pent-2-ene. Thus, three distinct alkene isomers are formed.	[1]
98	A	Option A is correct. The atmospheric window (8-14 micrometers) is normally transparent to outgoing Earth radiation. CFC C-F and C-Cl bond stretching vibrations absorb precisely within this window, acting as an extremely efficient thermal radiation trap.	[1]
99	D	Option D is correct. HFOs possess an alkene double bond (C=C). Unlike saturated CFCs or HFCs, the C=C bond is rapidly attacked by tropospheric .OH radicals within days, preventing them from reaching the stratosphere (ODP = 0) and giving an ultra-low global warming potential (GWP < 1).	[1]
100	B	Option B is correct. Boiling with NaOH(aq) hydrolyzes the chloroalkane to release Cl ⁻ ions. Acidifying with dilute HNO ₃ neutralizes unreacted hydroxide (preventing AgOH precipitation), and adding AgNO ₃ (aq) precipitates white silver chloride (AgCl), confirming the chloroalkane.	[1]
101	C	Option C is correct. Elimination of HBr from 2-bromobutane (CH ₃ -CH(Br)-CH ₂ -CH ₃) can occur in two directions: (1) abstraction of H from C1 yields but-1-ene (CH ₂ =CH-CH ₂ -CH ₃); (2) abstraction of H from C3 yields but-2-ene (CH ₃ -CH=CH-CH ₃). Because but-2-ene exhibits geometric stereoisomerism, it exists as two distinct stereoisomers: cis-but-2-ene and trans-but-2-ene. Total distinct alkenes = 1 + 2 = 3.	[1]
102	A	Option A is correct. 2-chloro-2-methylpropane is a tertiary haloalkane that ionizes almost instantaneously via the S _N 1 mechanism to form a stable 3° carbocation and free Cl ⁻ ions, forming a white precipitate of AgCl in under 10 seconds. Among primary haloalkanes, reaction proceeds via slow S _N 2: 1-iodobutane is faster than 1-bromobutane, while 1-chlorobutane has the highest bond enthalpy (340 kJ/mol) and forms a precipitate slowest (often taking > 15 minutes).	[1]
103	D	Option D is correct. The starting material has 2 carbons (bromoethane) while the final amine has 3 carbons (propan-1-amine). The carbon chain must be extended by one carbon. Step 1: Nucleophilic substitution with ethanolic KCN under reflux yields propanenitrile (CH ₃ CH ₂ CN). Step 2: Reduction of the nitrile using LiAlH ₄ in dry ether (or catalytic hydrogenation with H ₂ /Ni) yields propan-1-amine (CH ₃ CH ₂ CH ₂ NH ₂).	[1]
104	B	Option B is correct. For compound P, rate is proportional to [OH ⁻], which corresponds to a second-order rate law: Rate = k[substrate][OH ⁻], characteristic of the bimolecular S _N 2 mechanism favored by primary haloalkanes. For compound Q, rate is independent of [OH ⁻], corresponding to first-order kinetics: Rate = k[substrate], characteristic of the unimolecular S _N 1 mechanism favored by tertiary haloalkanes.	[1]
105	A	Option A is correct. The chlorine radical acts as a classic homogeneous catalyst. In step 1, it reacts with ozone to form ClO. and O ₂ . In step 2, ClO. reacts with an oxygen atom to regenerate the identical .Cl radical. Because it is continually regenerated, a single radical survives through thousands of cycles until rare radical termination occurs.	[1]
106	C	Option C is correct. The S _N 2 mechanism proceeds strictly via backside attack by the hydroxide nucleophile at 180° to the leaving chloride ion. This concerted attack forces complete stereochemical inversion (Walden inversion) of the chiral center, converting the pure (R)-enantiomer quantitatively into the pure (S)-enantiomer with optical activity.	[1]
107	B	Option B is correct. When a primary haloalkane reacts with ammonia, the primary amine formed (CH ₃ CH ₂ CH ₂ CH ₂ NH ₂) retains a nucleophilic lone pair on nitrogen and attacks unreacted haloalkane molecules, forming secondary amine (R ₂ NH), tertiary amine (R ₃ N), and finally the quaternary ammonium salt (R ₄ N ⁺ Br ⁻). All four species are present in the product mixture.	[1]
108	D	Option D is correct. HFC-134a (CF ₃ -CH ₂ F) contains only carbon, hydrogen, and fluorine atoms; it contains no chlorine or bromine atoms. Because it cannot generate chlorine or bromine free radicals upon photolysis in the stratosphere, its Ozone Depletion Potential is precisely zero.	[1]
109	A	Option A is correct. 2,3-dibromobutane has bromine atoms on C2 and C3. Elimination of the first HBr yields 2-bromobut-2-ene. Elimination of the second HBr removes the remaining hydrogen and bromine from C2 and C3, introducing a second pi bond to form the internal alkyne but-2-yne (CH ₃ -C#C-CH ₃).	[1]
110	C	Option C is correct. Molar mass of AgCl = 107.9 + 35.5 = 143.4 g/mol. Moles of AgCl precipitate = 2.87 g / 143.4 g/mol = 0.020 mol. Since 0.010 mol of the chloroalkane produced 0.020 mol of chloride ions (precipitated as AgCl), the mole ratio of Cl ⁻ to chloroalkane is 0.020 / 0.010 = 2. Thus, each molecule of the chloroalkane contains exactly 2 chlorine atoms.	[1]